Cold-Start Latency in Managed Vector Search: A Two-Week Measurement

Anonymous Author(s)
Submitted for blind review

Abstract—Managed vector indexes are usually described by their steady-state recall and throughput. We instead measure the first query after an idle period, the cold start, on one managed service over fourteen days. Across 2,480 cold starts the median latency was 780 ms, seven times the warm median, and the slowest decile exceeded four seconds. Cold-start cost scaled with index size but not with the number of replicas, which suggests the delay sits in loading index segments rather than in request routing. We report the method, the distribution, and the two configuration changes that removed most of the tail.

Index Terms—vector search, cold start, latency measurement, managed services

I. Introduction

Approximate nearest-neighbour indexes are memory resident by design. When a managed service scales a replica down, the next query must rebuild or reload the index before it can answer. Operators see this as an occasional slow query; users see it as a first interaction that feels broken. Published evaluations report warm percentiles, because warm measurement is easy to automate [1].

This paper asks a narrow question: how large is the cold-start penalty in practice, and what does it scale with? We measured one managed vector service for fourteen days, issuing the first query of every idle window and recording the service's own latency figures alongside client-side timing.

II. Method

We issued one query every fifteen minutes to a three-replica index of 4.2 million 768-dimensional vectors, and treated a query as cold when the preceding idle gap exceeded ten minutes. Warm queries ran in a separate stream against the same index so that both distributions come from the same days and the same data. We recorded client time to first byte, service-reported latency, and the replica that answered.

TABLE I
Latency by state, milliseconds, over fourteen days

State	Median	p90	Max
Warm	110	190	640
Cold	780	4,100	9,800

III. Results

Table I gives the two distributions. The cold median is 7.1 times the warm median, and the gap widens in the tail: the cold p90 is more than twenty times the warm p90. Cold starts were 3.1 per cent of queries but 41 per cent of the total time spent waiting.

Cold-start latency correlated with index size ($r = 0.71$) and not with replica count ($r = 0.08$), which points at segment loading rather than scheduling. Raising the minimum replica count from one to two, and enabling the service's keep-warm option, removed 86 per cent of cold starts above one second at a cost of 19 per cent more instance hours.

IV. Discussion

The measurement is limited to one service, one index shape and one region, so the numbers are not a ranking. What generalises is the shape: a heavy tail that steady-state benchmarks do not show and that a keep-warm setting can remove. We recommend that managed services report a cold-start percentile beside their warm figures, since the first query of a session is the one a person remembers.

References

[1] A. Example and B. Sample, "A survey of approximate nearest-neighbour benchmarks," Journal of Invented Systems, vol. 12, no. 3, pp. 210–219, 2025.

[2] C. Placeholder, "Notes on index reload behaviour in shared clusters," in Proc. Invented Conference on Systems Measurement, 2026, pp. 44–51.